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Electronics For You invites electronics professionals and industry experts to write articles on their area of expertise and interest from different perspectives

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Fig. 5: DJI Mavic Pro air foldable drone (Credit: www.dji.com/mavic)

inertial measurement unit (IMU), which comprises 3-axis gyroscopes and 3-axis accelerometers. An IMU tracks the path of the drone and stabilises its flight whenever the drone operates in windy conditions, harsh environments or is scouting in a radioactive area. Functions of an IMU are quite similar to those of a UAV; the only difference is the environments in which the vehicles operate.

Standards for sensors

IP500 standard ensures maximum interoperability to use and supports main industry norms. It uses and supports IPv6, IEEE802.15.4-2006, 6LoWPAN, IPSec, AES128 and BACnet. It fulfills all EN and VdS specifications for electronic devices for safety and security applications such as break-in protection, access control, fire protection, lightning and climate control, messaging location and emergency exits.

The commercial sector requires IP500 standard for installation and operation. It simplifies the servicing and maintenance of smart buildings and provides flexibility to use different components.

2700-2017-IEEE standard is a sensor performance parameter of sensor technologies with digital input/output (I/O) interfaces. It provides the framework for sensor performance specification, units, terminology, limits and condition with the intent to ease system integration and accelerate time-to-market.

Accelerometers that can withstand very high temperature environment are used for vibration testing of aircraft gas turbine engines, rocket propulsion systems and exhaust systems. UHT-12, a new crystal, is designed for more accurate and low noise measurements during large temperature variations. Data quality of sensors made with UHT-12 technology has improved as compared to ceramic crystal designs.

Measurements of force and pressure on the aircraft or rocket test models help engineers predict these on a full-scale aircraft to improve its aerodynamic performance. There are many types of wind tunnels designed for specific speed ranges to measure aircraft noise, impact penetration, rapidly fluctuating pressure and cavitation measurements.

Pressure accelerometers, microphones, force and torque sensors are used for numerous applications such as testing rocket motor combustion instability, gas turbine engines, cryogenic fuel pulsations, helicopter health and usage monitoring (HUMS), hydraulic pump evaluation, catalytic converter testing and brake testing.

New innovations like use of sensors, drones, robots, aircrafts and satellites have significantly improved the level of security and safety of defence armed forces. In addition, these innovations continuously monitor and reduce the risk of life at every stage. **EFY**